

For this term paper I chose the dataset "EEG Database", Donated to <https://archive.ics.uci.edu/datasets> on 10/12/1999, created by Henri Begleiter. The data was gathered to identify differences in neuroactivity, in relation to use of alcohol in the test subjects. The full original data consists of 122 test subjects, 61 alcoholic subjects and 61 subjects in control group. Each subject went through 120 trials where they were exposed to different stimuli. EEG, electroencephalogram, is a test that measures electrical activity in the brain. The EEG, measures neuroactivity using electrodes placed on subject's scalp, sampling at 256 Hz for 1 second. This means sampling is executed one second after stimulus and generates 256 observation points, T . The small dataset available on the site consists of one alcoholic and one control subject, showing 10 trials instead of 120.

Research plan is as follows: one test subject from the small dataset is well enough to reach the goal of this paper. Out of the ten trials for the subject, we can create an averaged data across time. Therefore it is important to remember that the data depicts the average response trajectory. This may also reduce noise as single outlier is less remarkable. From the small dataset I focus on one test subject and model dependence between activities in key parts of the brain. To construct a reasonably concise multivariate time series, I have picked three main electrodes (locations of data-gathering data from the brain) from 64 electrodes in the original data.

The selected three representative parts of the brain are all major functional brain regions: frontal (Fz), central (Cz), and parietal (Pz). Using these measurements, the objective is to create, and identify suitable model for a $T = 256$, $K = 3$ -dimensional observed time series. It is convenient to select from two groups of test subjects one that belongs in the control group to focus entirely on modelling instead of evaluating influence of alcohol in one test subject without control.

What we have now is a multivariate stochastic process of three components, Fz, Cz and Pz at time T ,

$T = 0, \dots, 255$ and $K = 3$.

$$\mathbf{Y}_t = [Fz_t, Cz_t, Pz_t]'$$

Here the variables can also be denoted as y_{1t} , y_{2t} and y_{3t}

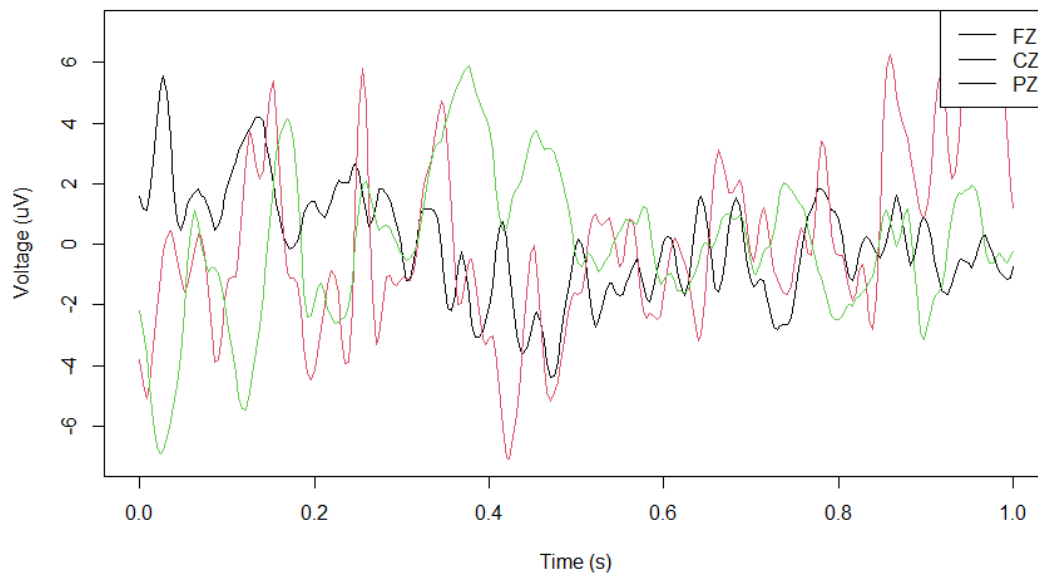


Figure1. Graphical presentation. Here (frontal) Fz=black , (central) Cz=red , (parietal) Pz=green

Figure1 displays EEG signals from the frontal, central, and parietal parts of the brain within one second after being engaged with stimuli. We can see wave-like oscillations in each of the samples, but as this phenomenon is very stable within each part the brain, not persistent or trending. No deterministic trend or systematic growth can be identified within the figure. Each series' mean is relatively plausibly around zero and variances also look roughly constant. Taking into account these qualities and while visual analysis can't prove stationarity, we can't reject the idea of these processes being stationary. These visual observations support the consideration of a Vector Autoregressive-model, VAR(p), to help present the potential cross-sectional dependencies.

To further picture the dependence between these three processes, and to guide lag-length selection for a VAR(p)-model, we continue to examine the autocorrelation structure of each univariate component series.

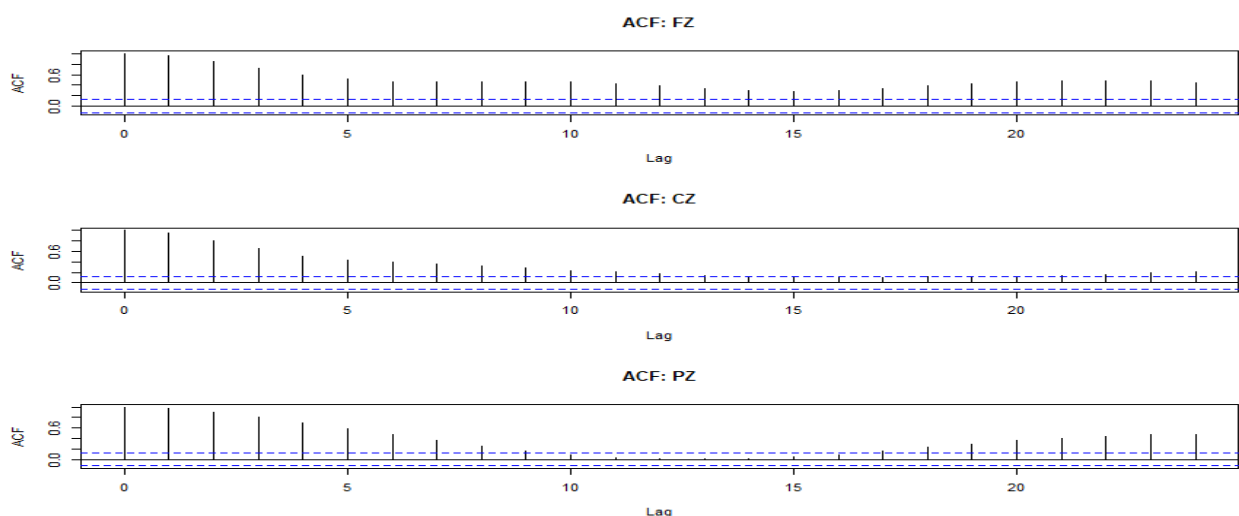


Figure2. Autocorrelation functions for Fz, Cz, Pz.

What we can immediately see from the autocorrelation functions of our components is that there is strong autocorrelation at short lags, this points to serial dependence as was earlier mentioned. For Fz the autocorrelation stays comfortably outside the confidence bands, while Cz only barely drops within. Overall, the autocorrelations appear to decay slowly, rather than drop to zero on short lags. After decay, each autocorrelation starts increasing in different capacities, most significantly Pz. This supports the oscillatory- patterns, which still seems stable and suitable for VAR(p)- model.

To determine the suitable lag length p , I apply the Akaike Information Criterion (AIC) and the Bayesian Criterion (BIC) using the VARselect procedure. Based on previous figures, at the same time a high suggested lag length may be expected but we also don't want to overparametrize our VAR- model. After setting maximum lag at 8, AIC suggested the maximum lag of eight and BIC suggested a lag of four. Even on higher maximum lags AIC suggested the maximum up to 14. Therefore we can comfortably move forward with $p=4$, as per suggested by BIC. It is promising that even a lag of 4 already captures extremely short-run neural dynamics but a satisfying amount of information.

Before forming our model frame, it is essential to make sure of stability through eigenvalues.

0.9548016	0.9054487	0.9054487
0.8771444	0.8728189	0.8728189
0.8279447	0.8279447	0.6671132
0.6671132	0.4294053	0.2093469

Figure3. Matrix of eigenvalues for Var(4)- model.

Evaluation of stability of the estimated VAR(4)-model can be executed by inspecting eigenvalues of the companion matrix, as in Figure3. All the roots are within unit circle (<1), with the largest modulus being 0.955. The VAR model shall be considered stable and is confirmed to satisfy the stationary condition.

We have made sure the conditions for VAR(p) are filled, except for serial non-correlation in residuals which will be checked in diagnostics.

For the estimated model $\mathbf{Y}_t = [\mathbf{Fz}_t, \mathbf{Cz}_t, \mathbf{Pz}_t]'$, we can also use notations $\mathbf{Y}_t = [Y_{1t}, Y_{2t}, Y_{3t}]$. In figure4 A_i , $i=1,2,3,4$, A_i are a 3×3 matrix of coefficients, which describe lag persistence, influences of past lags on current activity between the three regions in brains.

$$\begin{aligned}
\begin{pmatrix} Fz_t \\ Cz_t \\ Pz_t \end{pmatrix} &= \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} + A_1 \begin{pmatrix} Fz_{t-1} \\ Cz_{t-1} \\ Pz_{t-1} \end{pmatrix} + A_2 \begin{pmatrix} Fz_{t-2} \\ Cz_{t-2} \\ Pz_{t-2} \end{pmatrix} + A_3 \begin{pmatrix} Fz_{t-3} \\ Cz_{t-3} \\ Pz_{t-3} \end{pmatrix} + A_4 \begin{pmatrix} Fz_{t-4} \\ Cz_{t-4} \\ Pz_{t-4} \end{pmatrix} + u_t \\
Fz_t &= C_1 + a_{11}^1 Fz_{t-1} + a_{12}^1 Cz_{t-1} + a_{13}^1 Pz_{t-1} + \dots + a_{13}^4 Pz_{t-4} + u_{1t} \\
Cz_t &= C_2 + a_{21}^1 Fz_{t-1} + a_{22}^1 Cz_{t-1} + a_{23}^1 Pz_{t-1} + \dots + a_{23}^4 Pz_{t-4} + u_{2t} \\
Pz_t &= C_3 + a_{31}^1 Fz_{t-1} + a_{32}^1 Cz_{t-1} + a_{33}^1 Pz_{t-1} + \dots + a_{33}^4 Pz_{t-4} + u_{3t}
\end{aligned}$$

figure4. model specification sori

Where c is a constant and VAR model assumption that residuals u_t are white noise process with expected value of zero, furthermore

$$\begin{aligned}
E(u_t) &= 0 \\
E(u_t u_t') &= \Sigma u \\
E(u_t u_{t-h}) &= 0, \text{ when } h \text{ is nonzero.}
\end{aligned}$$

Each individual equation has it's own constant, lags, cross-channel lags and error term. This makes in total 39 parameters in the model, which is reasonable for $T=256$ and $K=3$.

The estimated coefficients indicate strong dependence to their own lags in all three equations, with only one exception (Fz_{13} for Fz surprisingly gives $p=0.761$). This is consistent with the large early-lag autocorrelations observed in the univariate ACFs and the strong short-run persistence of EEG signals. There are no significant dependencies for Cz but for Pz , almost every cross-factor helps predict activity in Pz even p -values lower than 0.005.

Granger-causality test goes further than just individual coefficients as formerly. By Granger-causality we examine whether lagged values of one variable would jointly provide improved predictive influence to the others.

Causing variable	Null hypothesis	F-test value	p-value	Verdict on H_0
Fz	Fz does not Granger-cause Cz & Pz	1.6695	0.1023	Do not reject
Cz	Cz does not Granger-cause Fz & Pz	2.2529	0.02219	Reject
Pz	Pz does not Granger-cause Fz & Cz	3.3779	0.0008166	Reject

table1. Granger-causality results.

The null hypothesis that Pz does not Granger-cause Fz and Cz was strongly rejected ($p < 0.001$), which means that past parietal neuroactivity does contain predictive information for both frontal and central regions. Similarly, the hypothesis that Cz does not Granger-cause the other parts of brain was rejected ($p = 0.022$), indicating a predictive role for the central region toward other regions. The null hypothesis that FZ does not Granger-cause the other regions could not be rejected ($p = 0.10$), indicating weaker evidence of predictive factor from the frontal region.

These results are evidence of dynamic dependence across regions of brain, with particularly strong short-run predictive effects originating from the parietal region.

Next in diagnostics, it's essential to test significance of auto and cross correlation in residuals to evaluate the adequacy of the estimated VAR- model. We can execute this using the Portmanteau test.

Model	Test lags	Degrees of freedom	Chi-squared test	p-value
VAR(4)	16	108	290.65	< 0.00001
VAR(6)	16	90	198.78	< 0.00001

table2. Portmanteau test for residual autocorrelation

The Portmanteau test rejects the null hypothesis of no residual autocorrelation. This means that higher-order dynamics may be present. Indicating, that the VAR(4) specification does not fully capture the dynamic structure of the EEG test. In comparison we have included in *table2*, how a model of higher dimension, VAR(6) model would endure the Portmanteau test. The extended version only gave a slightly better p-value but still easily forces to reject the null hypothesis. The Chi-squared test statistic reduced from 290.65 to 198.78, implying there was a positive change. Due to parsimony VAR(4) model is retained as a baseline specification.

There is left a remarkable amount of serial-correlation in the residuals, which points to the model being misspecified. regardless, we will go through with diagnostics.

Residual cross-correlation functions can be examined to study remaining cross-channel dependence.

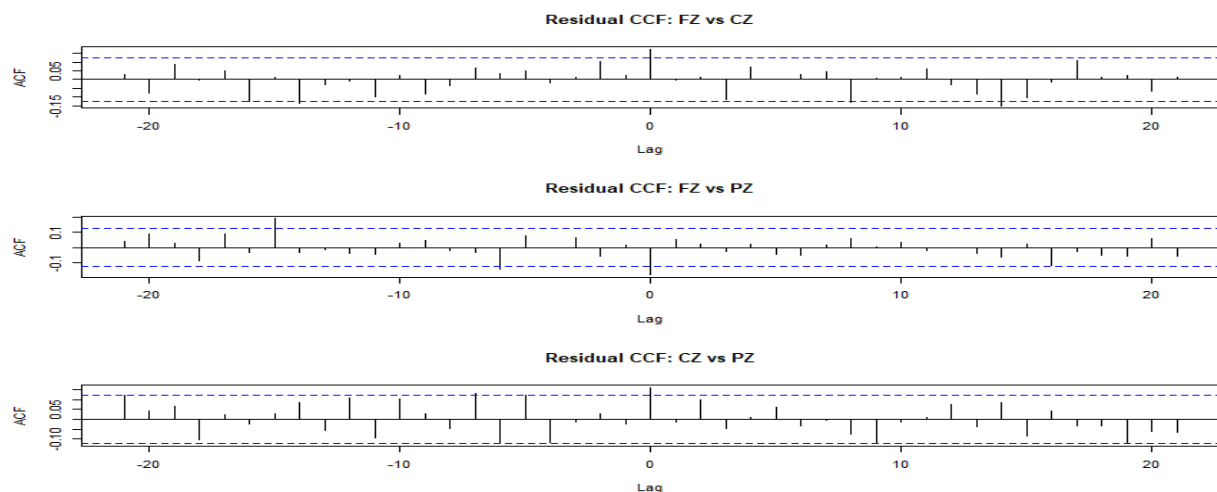


Figure5. Residual cross-correlation functions.

Despite the rejection in Portmanteau test, the residual cross-correlation functions appear well-behaved. The CCFs do not reveal significant misspecification remaining in the pairs of variables. Apart from a few isolated spikes at individual lags, most cross-correlations lie within the confidence bounds and do not display patterns that could indicate misspecification. This suggests that the VAR(4) specification captures the primary dynamic interactions between regions in EEG. The rejection of the Portmanteau test, or remaining inadequacy of VAR(4)-model, may be due to deeper serial dynamics rather than substantial unmodeled interactions.

Finally, to inspect squared residuals visually

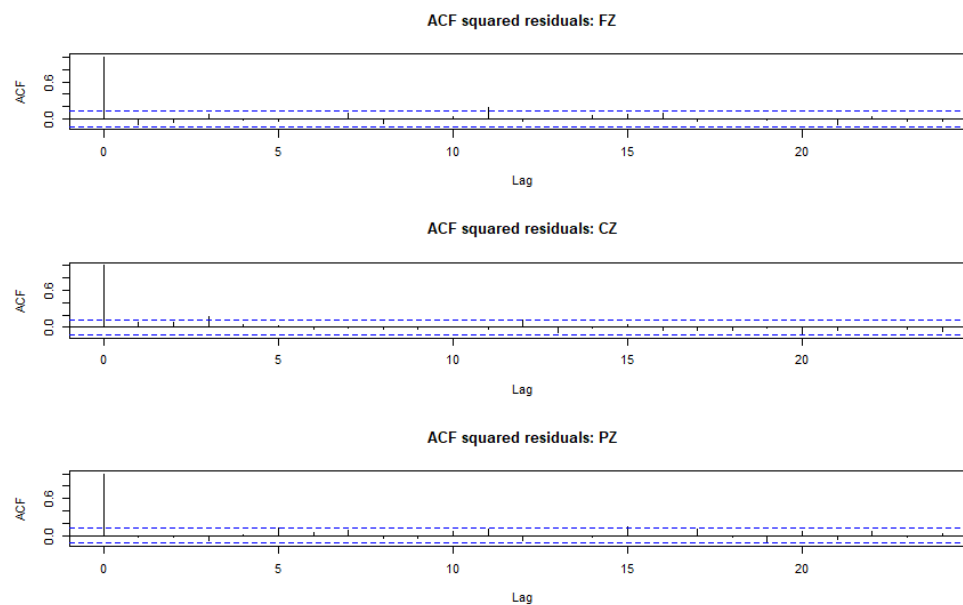


Figure6. ACF squared residuals

To assess potential ARCH effects, we make sure the squared residuals are within bounds and do not show systematic patterns. For all three equations, squared residual autocorrelations at non-zero lags lie within the confidence bounds, showing no evidence of conditional heteroskedasticity. This shows that there's no conditional heteroscedasticity in the residuals u_t , and the innovation variance appears stable over several lags. Along with results in residual normality, this further proves that the diagnostic problem lies in deeper serial dynamics.

Together with the residual normality results, this suggests that the main diagnostic concern of the VAR(4) model is remaining serial dependence rather than variance misspecification.

Conclusion

This study examined dynamic interdependence between frontal (FZ), central (CZ), and parietal (PZ) EEG signals using a multivariate timeseries-model. Given the detected potential interactions between regions of brain, a VAR-model provided the most convenient frame of model. After making sure the conditions of the study were appropriate, a VAR(4) model was fitted. This was found to be dynamically stable, with all eigenvalues lying inside

the unit circle. Granger-causality tests provided further evidence of directional predictive relationships. Parietal activity strongly predicted both frontal and central activity, while central activity also proved to have predictive influence. Residual diagnostics supported the stability of the model and that innovations were normally distributed, to a satisfactory extent. Regardless of promising results, Portmanteau test revealed that some residual serial correlation remained in the model, which may indicate to the VAR(4)- model not capturing the full structure of EEG-signals. Since this analysis was only based on one subject, and reduced to three core pieces of the EEG system, these results should be interpreted with caution.

```

combined <- dplyr::bind_rows(list, .id = "run")

avg_data <- combined %>%
  dplyr::group_by(time) %>%
  dplyr::summarise(
    FZ = mean(FZ),
    CZ = mean(CZ),
    PZ = mean(PZ)
  ) %>%
  dplyr::ungroup()

#The stochastic process
Y <- avg_data %>% dplyr::select(FZ, CZ, PZ)
Y <- scale(Y, center = TRUE, scale = FALSE)

#graphical presentation
t <- seq(0, 1, length.out = nrow(Y)) #to represent 1 second of sampling
matplot(t, Y, type="l", lty=1, xlab="Time (s)", ylab="Voltage (uV)")
legend("topright", legend=c("FZ","CZ","PZ"), lty=1)

#cross correlations
ccf(Y[, "FZ"], Y[, "CZ"], main = "CCF: FZ vs CZ")
ccf(Y[, "FZ"], Y[, "PZ"], main = "CCF: FZ vs PZ")
ccf(Y[, "CZ"], Y[, "PZ"], main = "CCF: CZ vs PZ")

#information criteria, make sure no trivial time dependence
par(mfrow=c(3,1))
acf(Y[,1], main="ACF: FZ")
acf(Y[,2], main="ACF: CZ")
acf(Y[,3], main="ACF: PZ")
par(mfrow=c(1,1))

#to select lag for VAR model
lag_sel <- VARselect(Y, lag.max = 8, type = "const")
lag_sel$criteria
lag_sel$selection
#BIC suggests p=4
p <- lag_sel$selection["SC(n)"]
fit <- VAR(Y, p = 4, type = "const")
summary(fit)
#stability of the model, eigenvalues
eig<-roots(fit)
eig_matrix <- matrix(eig, nrow = 4, byrow = TRUE)
eig_matrix
###granger causality
causality(fit, cause = "FZ")
causality(fit, cause = "CZ")
causality(fit, cause = "PZ")

```



```

#Diagnostics
serial.test(fit, lags.pt = 16, type = "PT.adjusted")
#we try a higher dimension var-model
fit6 <- VAR(Y, p = 6, type = "const")
serial.test(fit6, lags.pt = 16, type = "PT.adjusted")
#
normality.test(fit)           #normality of residuals
arch.test(fit, lags.multi = 5) #check heteroskedasticity
stability(fit)               #stability check
plot(stability(fit))         #roots plot

res <- resid(fit)
ccf(res[, "FZ"], res[, "CZ"], main = "Residual CCF: FZ vs CZ")
ccf(res[, "FZ"], res[, "PZ"], main = "Residual CCF: FZ vs PZ")
ccf(res[, "CZ"], res[, "PZ"], main = "Residual CCF: CZ vs PZ")

par(mfrow=c(3,1))
acf(res[, "FZ"]^2, main="ACF squared residuals: FZ")
acf(res[, "CZ"]^2, main="ACF squared residuals: CZ")
acf(res[, "PZ"]^2, main="ACF squared residuals: PZ")
par(mfrow=c(1,1))

```